

AudioPlatform

for Teensy 4.1

Getting Started Guide



Revision 1.0.1, July 31, 2025

Donald Swearingen and ProtoSupplies

Welcome

First and foremost, AudioPlatform for Teensy 4.1 is a hardware platform for experimentation and development of simple-to-sophisticated, stand-alone, audio systems and applications, built within and upon the Teensy4.1/TeensyDuino/Arduino environment.

AudioPlatform is a response to the difficulty that many potential creators of Teensy-based audio systems face: the challenge of acquiring or themselves building some sort of hardware, often a “DIY” PCB board and list of components, to accommodate their projects and explorations. Many of those with such interests simply lack the skills or resources to complete this process; and the general lack of availability of off-the-shelf hardware suitable to their projects restricts their options to the use of breadboards and/or simply-wired “perf” boards to do so, not to mention creating modern PCB-based devices with surface mounted components.

This was pretty much the case for me, as well. Although I’m a life-long professional musician and composer, I’ve also over the years acquired considerable skills in software and electronic systems development, creating software and hardware components and systems in support of numerous live musical and artistic projects. Most of that work had also involved the use of powerful computers.

AudioPlatform aims to address that deficit by offering a system that includes a significant set of resources for audio applications. These include both hardware and software components.

What's In the Box

- AudioPlatform Unit
- 7.5V AC/DC Adapter
- 32GB MicroSD Card
- MicroSD to SD adapter
- USB-C cable
- Tilting base
- Hardcopy getting started guide?

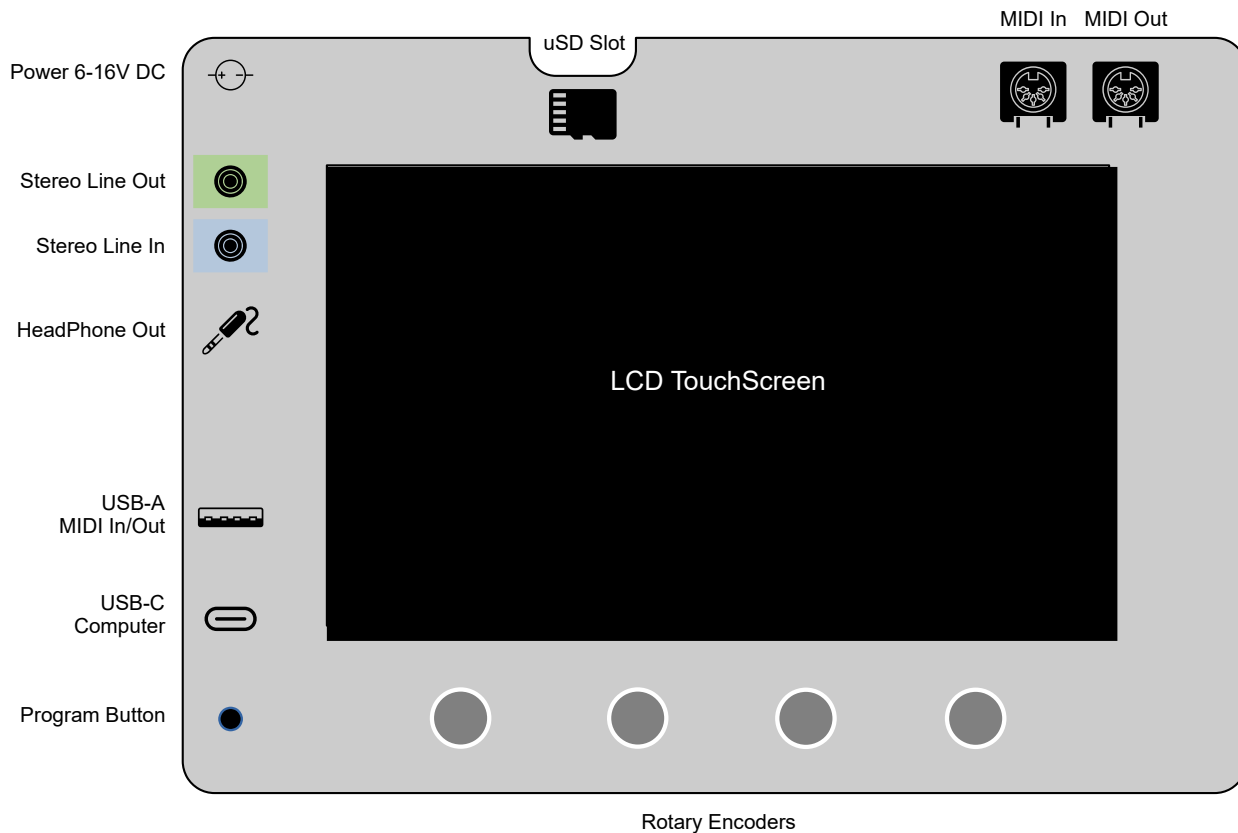
microSD Card

Although the AudioPlatform application can be operated without the presence of a microSD card (generally referred to in this document as the SD Card) in the Teensy 4.1 card slot, there are a number of features of the system that are dependent on the presence of the card and its contents. These include:

1. Access to Stereo WAV (".wav") audio files located on the SD card for playback by the SD Wav Player
2. Access to banks of Monaural (Mono) RAW (".raw") audio files to be transferred to the internal Serial Flash storage for playback by the Flash Raw Player
3. Access to user Preset Banks (".pbnk") files stored on the card
4. Access to user-defined Voice Banks (".syx") for playback using the FM Synthesizer

System Setup and Verification

Input/Output Connections



Connection Details

Power 6-16V DC

Power is delivered through the DC input connector from a standard 9VDC wall adapter with a center-positive (+), 2.1mm female connector. Note that power supplied through the Teensy 4.1 USB-C connection to a computer is disabled—AudioPlatform can only be powered from the DC input.

Ken: Normally ship with a 7.5V adapter

Stereo Line In/Out

Stereo Line inputs and outputs are female 3.5mm TRS connections.

Ken: Maybe worth calling out SGTL5000 audio subsystem

HeadPhone Out

The headphone output is a 3.5mm TRS female connector.USB-A Connector

USB-A Connector

The USB-A connector is used for MIDI input from a MIDI keyboard or controller, and can supply power to a connected MIDI device.

Ken: Up to a maximum of 500mA. Higher current devices need to be externally powered.



In the first release of the AudioPlatform application (apApp), only MIDI input is supported. Support for MIDI output, and multiple input devices via a USB Hub are planned for future versions.

USB-C Connector

The USB-C connection is used exclusively for connection of AudioPlatform to a computer and supports both the Arduino IDE, and Serial IO.



Support for MIDI and Audio IO via the USB-C connection is planned for a future release of the AudioPlatform application.

Program Button

The small program button is employed in the process of downloading a pre-compiled executable (".exe") image to the AudioPlatform's Teensy 4.1, when using the Teensy Loader application (an independent component of the Arduino/Teensy environment which allows downloading independent of whether or not the Arduino IDE is active).

MicroSD Card Slot

The microSD (uSD) slot accepts a standard microSD card. Though AudioPlatform can start up and perform many of its functions without the presence of a uSD card, there are numerous features and capabilities of the apApp that depend on and require a card to be present. The types and compatibilities of various uSD media are discussed further in **SECTION REFERENCE**.

Ken: 32GB was the limitation. Now in theory can support up to 2TB but I haven't tried anything other than 32GB which is what I ship with

<https://forum.pjrc.com/index.php?threads/microsd-card-max-size-in-teensy-4-1.67242/>

Paul StofReggen 5/19/21

You need to use SdFat or the latest 1.54 betas (where SD is now just a thin wrapper for SdFat) for cards larger than 32GB.

If you use Teensyduino 1.53 or older, the old Arduino SD library supports only up to 32GB. If you're not sure which version you have, in Arduino click Help > About (or Teensyduino > About on Macintosh) to see which version is installed.

Paul StofReggen 10/17/22

Use FAT for 32GB and smaller cards, or exFat for larger than 32GB.

Theoretically up to 2TB should work, but as far as I know 1TB is the largest card anyone has tried. The largest card I own is 512GB. It works fine.

KurtE 10/17/22

Note: with SD library usage of SDFat, there are different classes you can setup to use: FatLib – which does: Fat16, Fat32 and optionally Fat12 ExFatLib – which does ExFat FsLib – which is setup to use Fat or ExFat depending on the card.

Our SD Library uses the FsLib (at least for most teensy boards)

We have format code built in, including in some of our sketches. However, you should be able to use any formatter on your PC.

I know at least at one point the developer of SDFat recommended the formatter

from: <https://www.sdcard.org/downloads/formatter/> Which should initialize and format the cards to the SD standards...

MIDI DIN In/Out

MIDI input is also possible on the 5-pin DIN connector (original MIDI standard still used with numerous current MIDI devices).



Support for MIDI output, and simultaneous use of both the MIDI DIN connection, and the USB-Host MIDI IO connection are planned for apApp.

Rotary Encoders

The 4 rotary encoders are an important and integral component of the AudioPlatform design. In the AudioPlatform application, the encoders are used to enter settings and parameters of the various devices implemented in the application. Detailed descriptions of encoder usage are outlined in **SECTION REFERENCE**.

LCD Screen

Ken: Probably worth also calling out 7" RA8875 on LCD

Getting Started Guide

Section/Topic Ideas

- Welcome
- What's in the Box
- Setup and Verification
 - Power
 - Audio Out
 - MIDI Controller
 - Computer Connection
- Quick Overview
 - Players
 - Efx and MIDI Devices
 - Presets and Parameters
 - Screens and Menus
 - Keyboard Mapping
- Play Around and Have Some Fun
 - Presets Live: Load and Select Presets
 - Change Device Settings
 - Saving Changes
- Appendix
 - SD and Flash Data Contents in Shipped System
 - SD File Management and Maintenance
 - References and Links

Deliverables

microSD Card

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- Access to Stereo WAV (".wav") audio files located on the SD card for playback by the SD Wav Player
- Access to banks of Monaural (Mono) RAW (".raw") audio files to be transferred to the internal Serial Flash storage for playback by the Flash Raw Player
- Access to user Preset Banks (".pbk") files stored on the card
- Access to user-defined Voice Banks (".syx") for playback using the FM Synthesizer

Application Versions and Options

As discussed in detail in the *AudioPlatform User's Manual* and the *AudioPlatform Application and Programmer's Reference*, there are a number of application options that can be configured only by making modifications to the application source-code, and recompiling (rebuilding the application using the Audio IDE). However, so that users may explore and use the application without the need to set up the Arduino IDE and application software environment and carry out this process, the system is shipped with a number of pre-built "EXE" modules that can be easily downloaded into AudioPlatform without going through this process.

The 2 most important selections for getting started are 1) the selection of a generic or specific MIDI controller, and 2) whether the controller is to be connected to AudioPlatform via a USB connection, or via a standard MIDI "DIN" connection (5-pin round connector on both ends).

Default Options

- Generic MIDI Controller
- MIDI Connection via USB

Alternate Options

Alternate options can be employed both for the MIDI controller and the controller connection. Separate, installable EXE files are available for download from the link

<https://www.UrlLinkForAlternateEXEs.com>

Included are EXE modules for the following setups:

- Generic MIDI Controller, MIDI connection via USB
- Generic MIDI Controller, MIDI connection via DIN input
- Arturia Minilab3 MIDI Controller, MIDI connection via USB
- Arturia Minilab3 MIDI Controller, MIDI connection via DIN input
- Novation LaunchKey 25 MIDI Controller, MIDI connection via USB. *Note: LaunchKey 25 only includes a USB MIDI output. DIN connections are not available.*
- Novation LaunchKey 49 MK3 MIDI Controller, MIDI connection via USB
- Novation LaunchKey 49 MK3, MIDI connection via DIN input

Installing Alternate Options

Teensy Loader Application

Pre-built Teensy EXE executable files can be easily downloaded to AudioPlatform (or any compatible Teensy device connected to a computer via its standard USB connection) using the *Teensy Loader* application. Versions of Teensy Loader are available for Windows, MacOS, and Linux at the following locations:

Windows

https://www.pjrc.com/teensy/loader_win10.html

MacOS

https://www.pjrc.com/teensy/loader_mac.html

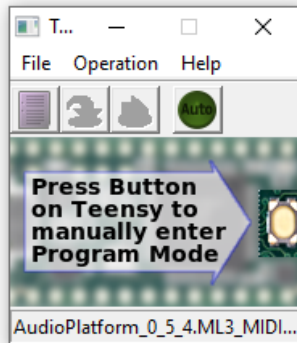
Linux (Ubuntu)

https://www.pjrc.com/teensy/loader_linux.html

Using Teensy Loader

Once you've downloaded the Teensy Loader and a suitable alternate EXE for AudioPlatform:

1. double click on the loader (Teensy.exe) to display its small screen:



2. Click on the File menu
3. Navigate to the location of the EXE to be installed and click to select (the name of the selected file will be displayed at the bottom of the loader window, as above)
4. Press the PROGRAM button at the bottom left and beneath the top panel of AudioPlatform to begin the download.

Once the download is completed, AudioPlatform will restart and display its Main Menu screen.



Among the options on the list for inclusion in a future version of the apApp are run-time selection of the MIDI controller and the connection port (USB or MIDI). In the case of the port connection, the intention is that, in the future, both ports will be active, although only a single controller may be recognized (another potential addition is the ability to use more than one controller, and perhaps a USB hub).

microSD Contents

Stereo WAV Files

Bank 0

000-SoundFile.wav
001-SoundFile.wav
002-SoundFile.wav
003-SoundFile.wav
004-SoundFile.wav
005-SoundFile.wav
006-SoundFile.wav
007-SoundFile.wav
008-SoundFile.wav
009-SoundFile.wav
010-SoundFile.wav
011-SoundFile.wav
012-SoundFile.wav
013-SoundFile.wav
014-SoundFile.wav
015-SoundFile.wav
016-SoundFile.wav
017-SoundFile.wav
018-SoundFile.wav
019-SoundFile.wav
020-SoundFile.wav
021-SoundFile.wav
022-SoundFile.wav
023-SoundFile.wav
024-SoundFile.wav

Bank 1

100-SoundFile.wav
101-SoundFile.wav
102-SoundFile.wav
103-SoundFile.wav
104-SoundFile.wav
105-SoundFile.wav
106-SoundFile.wav
107-SoundFile.wav
108-SoundFile.wav
109-SoundFile.wav
110-SoundFile.wav
111-SoundFile.wav
112-SoundFile.wav
113-SoundFile.wav
114-SoundFile.wav
115-SoundFile.wav
116-SoundFile.wav
117-SoundFile.wav
118-SoundFile.wav
119-SoundFile.wav
120-SoundFile.wav
121-SoundFile.wav
122-SoundFile.wav
123-SoundFile.wav
124-SoundFile.wav

Stereo WAV Files (continued)

Bank 2

200-
201-
202-
203-
204-
205-
206-
207-
208-
209-
210-
211-
212-
213-
214-
215-
216-
217-
218-
219-
220-
221-
222-
223-
224-

Bank 3

300-
301-
302-
303-
304-
305-
306-
307-
308-
309-
310-
311-
312-
313-
314-
315-
316-
317-
318-
319-
320-
321-
322-
323-
324-

Mono RAW Files

Bank 0

000-
001-
002-
003-
004-
005-
006-
007-
008-
009-
010-
011-
012-
013-
014-
015-
016-
017-
018-
019-
020-
021-
022-
023-
024-

Bank 1

100-
101-
102-
103-
104-
105-
106-
107-
108-
109-
110-
111-
112-
113-
114-
115-
116-
117-
118-
119-
120-
121-
122-
123-
124-

Mono RAW Files (continued)

Bank 2

200-
201-
202-
203-
204-
205-
206-
207-
208-
209-
210-
211-
212-
213-
214-
215-
216-
217-
218-
219-
220-
221-
222-
223-
224-

Bank 3

300-
301-
302-
303-
304-
305-
306-
307-
308-
309-
310-
311-
312-
313-
314-
315-
316-
317-
318-
319-
320-
321-
322-
323-
324-

SD Card Usage and Management

Although the presence of an SD card is not required to use many of the basic features of apApp, the full extent of application capabilities is only possible when an SD card is present in the Teensy 4.1 microSD slot.

Because of this and various other considerations, it is considered fundamental to the full benefits of AudioPlatform that the contents of the SD card be managed in a well-organized and careful manner.

Other than the capability to name, store, and load preset banks to/from the sdCard, apApp makes no provision of sdCard file and directory management. It is thus necessary that users organize and manage soundfile resources and SD card contents on a PC (Windows, Mac, Linux) using various available applications and/or utilities. The renaming of a collection of audio files to conform to the naming requirements for SD card audio files in apApp is perhaps the primary issue in this connection,

and it is necessary to use some appropriate means of doing so. There are numerous paid and free applications that can be used for this purpose. Examples include:

PC-Based File Management Utilities

Windows

- Microsoft PowerToys PowerRename
- Advanced Renamer
- Bulk Rename Utility
- File Renamer
- Metamorphose 2

MacOS

- Finder's Built-in Feature
- Transnomino
- NameChanger
- Metamorphose 2

Linux

- GPRename
- pyRenamer
- Thunar
- Metamorphose 2

SD Card Format

SD Card Content

File Naming

In order to be recognized in apApp, WAV and RAW audio filenames must follow a specific convention that indicates which of the 4 sample banks the file is to be located in, and which MIDI note number (0-99, 100-127 ignored) will play the sample.

The convention dictates that the sample filename must begin with a 3-digit number (leading 0s required) followed by a dash '-' and the filename, and ending with the suffix .wav or .raw as the case may be. The 3 digits indicate both the bank number:

- 000-099 Bank 0, MIDI notes 0-99

- 100-199 Bank 1, MIDI notes 0-99
- 200-299 Bank 2, MIDI notes 0-99
- 300-399 Bank 2 MIDI notes 0-99

Examples

Bank 0

- 000-Note0_Sample.wav
- 060-MiddleC_Sample.wav
- 099-Note99_Sample.wav

Bank 2

- 250-SoundEffect_01.wav
- 260-SoundEffect_02.wav
- 299-HighVoice.wav

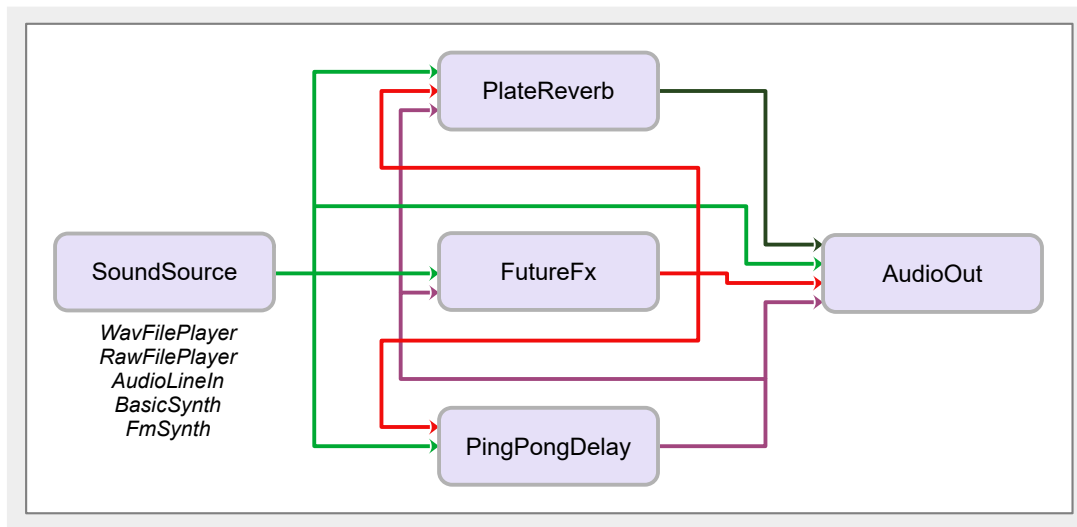
The same conventions are used for RAW audio file banks stored in non-volatile Serial Flash.

Filename Management

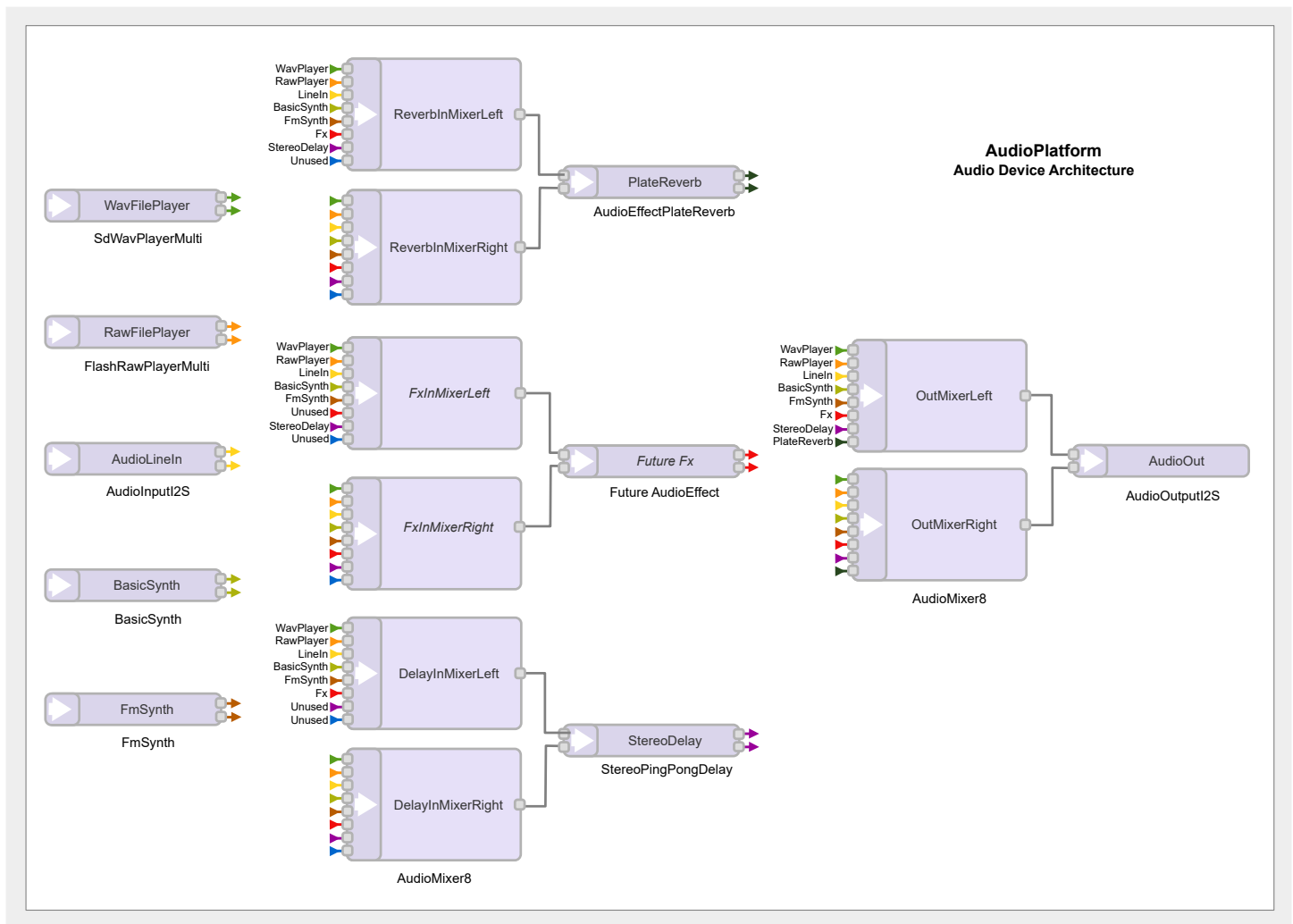
In general, the source of most user-supplied audio WAV and Raw files will be

Audio Architecture

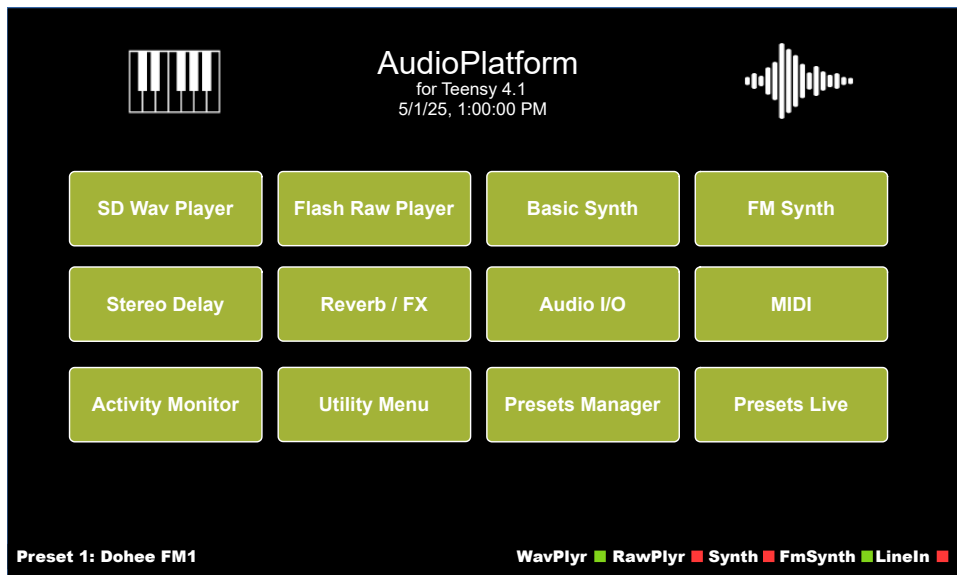
Signal Flow



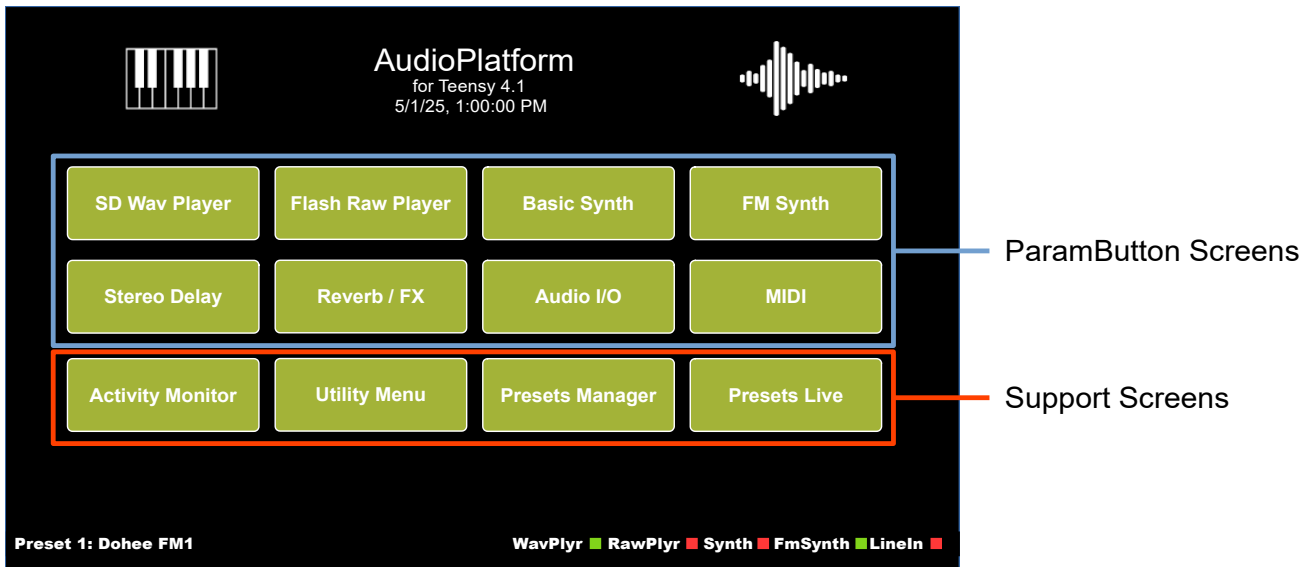
Connection Detail



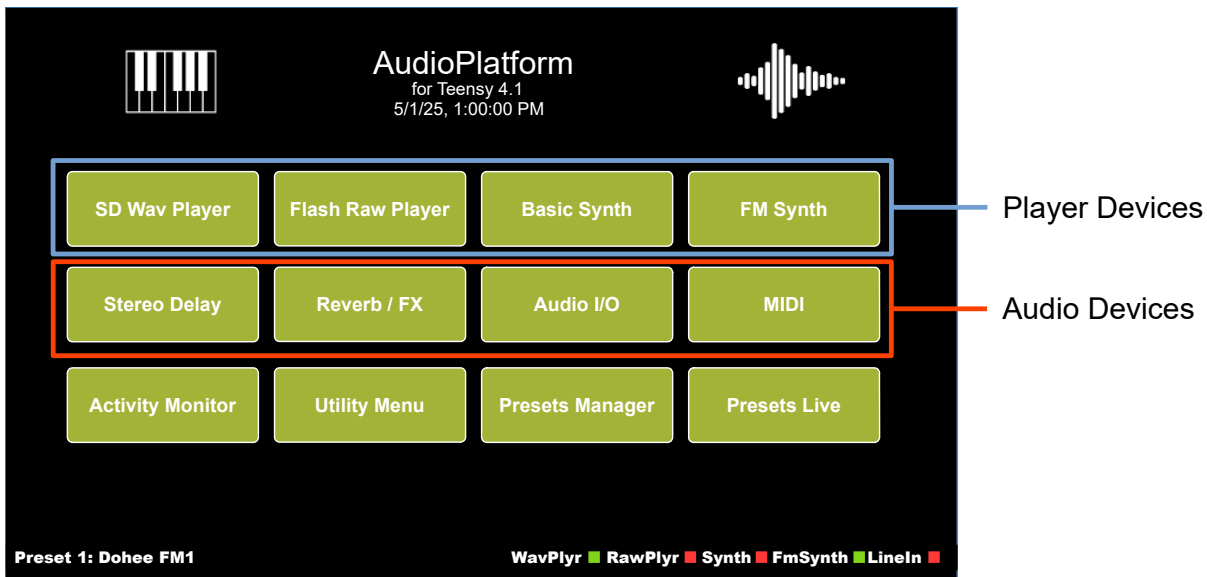
Main Menu



Main Menu Screen Types



Player and Audio Devices



Player Devices

Player Devices occupy the topmost row of the Main Menu screen. apApp features 4 players, which play audio media in response to inputs from an external MIDI device, and each of which operates on a separate media type:

- The *SD Wav Player* plays Stereo WAV formatted files located on the Teensy 4.1 SD card

- The *Flash Raw Player* plays Monaural (single channel, headerless) audio files located in the AudioPlatform's 128 MB non-volatile Flash Memory
- The *Basic Synth* is an implementation of a basic digital audio synthesizer with 12-voice polyphony
- The *FM Synth* is a Teensy Audio System implementation of the micro-Dexed synthesizer which is an emulator of the classic DX7 synthesizer, featuring 8-voice polyphony

Audio Devices

Audio Devices occupy the middle row of the Main Menu screen. apApp features 4 audio devices, which play various roles in the apApp audio architecture:

- The *Stereo Delay* device implements a stereo “ping-pong” digital delay, with up to 3 seconds of delay per channel, and both individual channel and cross-channel feedback paths
- The *Reverb / EFX* device features a Plate Reverberation unit, and a second effects unit pathway intended for addition of a 3rd audio effect in future apApp releases.
- The *Audio IO* device provides for control of the levels and effects settings of the system audio paths, with settings for Input and Output levels, effects send levels, and other audio system parameters
- The *MIDI* device features settings related to a connected MIDI control device, such as MIDI channel and destination player settings

Support and Utility Devices

Support Devices occupy the bottom row of the Main Menu screen, and serve as a gateway to additional screens and menus that perform various system and user support functions.

- The *Activity Monitor*
- The *Utility Menu*
- The *Presets Manager*
- The *Presets Live* device

Screens and Settings

Screen Icons

Screen icons (*buttons*) appear on all screens, and serve to allow the user to navigate among application screens (*Navigation Icons*), and to make selections or perform actions on the currently displayed screen (*Action Icons*).

Navigation Icons

Home



This button appears in the upper-right corner of all level 2 screens (screens one level below the main menu). Touching the button returns the user to the Main Menu screen.

Back



This button appears in the upper-right corner of level 3 and greater screens (screens at least two levels below the Main Screen).

Enter



This button may appear in various screens to provide access to a lower-level screen that is only accessible from the current screen.

Exit



This button appears in screens accessed using the Enter button, allowing the user to return to the screen which invoked the currently displayed screen.

Action Icons

All Notes Off



This button appears at the top of the Presets Live screen. When pressed, all notes currently playing are silenced. (Note: this function can also be assigned to a button or other control on a connected MIDI keyboard or controller).

Ignore Note Off



This button appears at the top of the Presets Live screen. When pressed, all notes currently held or played thereafter continue to play, ignoring key or sustain pedal releases. (Note: this function can also be assigned to a button or other control on a connected MIDI keyboard or controller).

Save Changes



This button appears at the top of all ParamButton screens. The text label of the button “Save Changes” normally appears in white. If any parameter changes are made to the displayed device using a rotary encoder, the text changes to red “**Save Changes**”, and remains so until the button is touched, at which point it returns to its normal white appearance. This process is discussed in detail in the **SECTION REFERENCE**.

Copy Settings



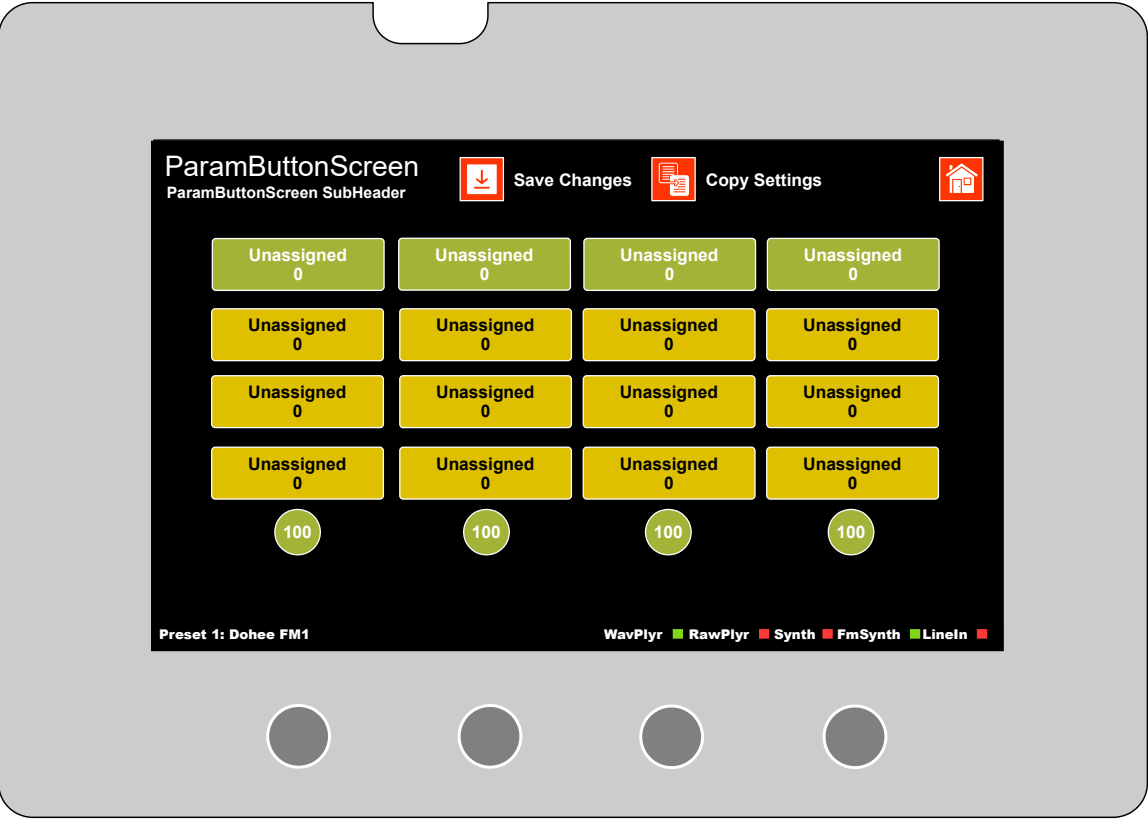
This button appears on all ParamButton screens next to the Save Changes button. When pressed, it displays a screen with buttons representing the presets currently loaded in the *Import* presets bank. When one of the import preset buttons is touched, the device settings of the import device are copied to the currently displayed device. This process is discussed in detail in the **SECTION REFERENCE**.

Scroll Up Down



These buttons appear on screens that display lists of files or other names either for review or selection of items (for example, a preset bank file on the SD card). When simply displaying lists, the screen will scroll up or down repeatedly until the button is released. When displaying items for selection, repeated touches are required to scroll the list.

ParamButton Screens



Player Device Screens

Player Device Common Settings

- **Player Status**
- **Player Level**
- **Player → Delay**
- **Player → Reverb**
- **Note Shift**
- **Note Min**
- **Note Max**
- **Pan Type**
- **Pan Parameter**

PitchToMIDI Simulation for WAV and Raw Players

Simultaneous Playback

1. In Ableton, generate MIDI data for the WAV sample in question.
2. Use the MIDI data to create a second WAV sample that plays an instrument using the MIDI data.
3. Output both WAV samples to drive, with similar names.
4. Load the 2 WAV samples onto SD (or Serial Flash) and play the samples simultaneously from the MIDI keyboard to simulate synchronization.

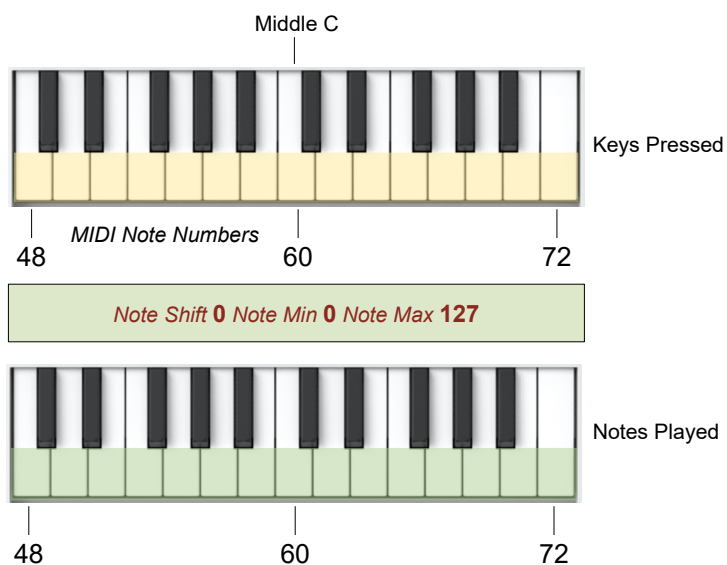
Variations

- Combine both original and MIDI sample in same stereo WAV file (hard LR, or mixed)
- (Best) Add a built-in Pitch to MIDI converter to AudioPlatform

Player Keyboard Mappings

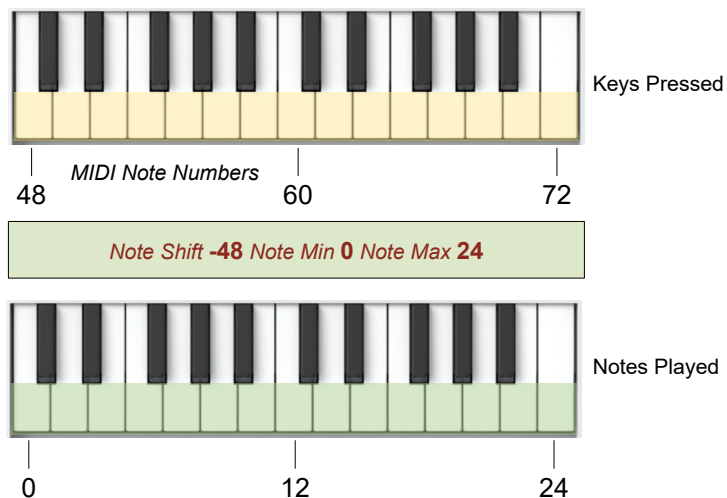
Among the device parameters of each Player Device are 3 settings that determine which keys on the MIDI keyboard will play which notes on the player's designated audio device.

Default Mapping

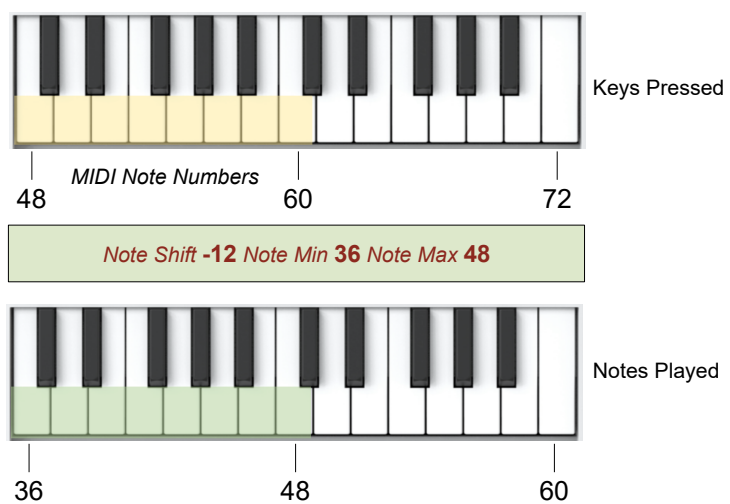


Shifted and Limited Mappings

All Notes Shifted Left 4 Octaves



12 Notes Shifted Left



WAV Player

WAV File Player

Teensy4.1 SD WAV Files

 Save Changes  Copy Settings 

Player Status Enabled	Player Level 0.0 dB	Player→Delay 58 %	Player→Reverb 80%
Sound Bank 0	Note Shift -12	Note Min 0	Note Max 99
Pan Type Center	Pan Parameter 63.0	Unassigned 0	Unassigned 0
1	0.5	1	1

Preset 1: Dohee FM1

WavPlyr  RawPlyr  Synth  FmSynth  LineIn 

RAW Player

Raw File Player

Serial Flash Raw Audio Files

 Save Changes  Copy Settings 

Player Status Disabled	Player Level -9.5 dB	Player→Delay 0 %	Player→Reverb 0%
Sound Bank 0	Note Shift 0	Note Min 0	Note Max 99
Pan Type Center	Pan Parameter 63.0	Unassigned 0	Unassigned 0
1	0.5	1	1

Preset 1: Dohee FM1

WavPlyr  RawPlyr  Synth  FmSynth  LineIn 

Basic Synthesizer

Basic Synthesizer
12 Voice Polyphony

 Save Changes  Copy Settings 

Synth Status Disabled	Synth Level 0.0 dB	Player→Delay 0 %	Player→Reverb 0%
Synth Type AM	Note Shift 0	Note Min 0	Note Max 99
Attack 0 ms	Decay 0 ms	Sustain 0 %	Release 0 ms
Mod Freq 0 Hz	Mod Amt 80 %	Pan Type Center	Pan Parameter 63.0
1	0.5	1	1

Preset 1: Dohee FM1

WavPlyr  RawPlyr  Synth  FmSynth  Lineln 

FM Synthesizer

Basic Synthesizer
12 Voice Polyphony

 Save Changes  Copy Settings 

Synth Status
Disabled

Synth Level
-8.5 dB

Player→Delay
20 %

Player→Reverb
60%

 Voices

Voice Name
Lute

Note Shift
14

Note Min
0

Note Max
127

PitchBend Range
1 Semitones

Pan Type
Center

Pan Parameter
63.0

Unassigned

1

0.5

1

1

Preset 1: Dohee FM1

WavPlyr  RawPlyr  Synth  FmSynth  Lineln 

FM Synthesizer Voice Banks

FM Synth Voice Banks
Select Bank and Voice

Voice

BRASS 1

BRASS 2

BRASS 3

STRINGS 1

STRINGS 2

STRINGS 3

ORCHESTRA

PIANO 1

PIANO 2

PIANO 3

E.PIANO 1

GUITAR 1

GUITAR 2

SYN.LEAD 1

BASS 1

BASS 2

E.ORGAN 1

PIPES 1

HARPSICH 1

CLAV 1

VIBE 1

MARIMBA

KOTO

FLUTE 1

ORCH-CHIME

TUB BELLS

STEEL DRUM

TIMPANI

REFS WHISTL

VOICE 1

TRAIN

TAKE OFF

Voice Bank

rom1a

rom1b

rom2a

rom2b

rom3a

rom3b

rom4a

rom4b

Ritch1

Ritch2

User1

User2

User3

User4

User5

User6


FM Synth

FM Synthesizer Emulator

synth_dexed is a port of the Dexed sound engine (<https://github.com/asb2m10/dexed>) as library for the Teensy-3.5/3.6/4.x with an audio shield.

Dexed ist heavily based on <https://github.com/google/music-synthesizer-for-android>

synth_dexed.cpp/h used in AudioPlatform is from:

Arduino Resources\Third Party Code\microdexed-touch\third-party\Synth_Dexed\src\

PC/Mac Editor

<https://asb2m10.github.io/dexed/>



Audio Device Screens

Stereo Delay

Audio IO Settings

Audio In/Out/Codec

 Save Changes  Copy Settings 

LineIn Status Disabled	LineIn Level 1.33 Volts p-p	LineOut Level 2.83 Volts p-p	Phone Volume 80 %
LineIn→Delay 0 %	LineIn→Reverb 100 %	Unassigned 0	Unassigned 0
1	1	1	1

Preset 1: Dohee FM1

WavPlyr  RawPlyr  Synth  FmSynth  LineIn 

Reverb and Special EFX

Effects Settings

Reverb and Future Fx

 Save Changes  Copy Settings 

Reverb Size 80	Reverb Diffusion 70 %	Reverb Level 0.0 dB	UnAssigned 0
Unassigned 0	Unassigned 0	Unassigned 0	Unassigned 0
0.5	1	0.5	1

Preset 1: Dohee FM1

WavPlyr  RawPlyr  Synth  FmSynth  LineIn 

Audio IO

Audio IO Settings

Audio In/Out/Codec

 Save Changes  Copy Settings 

LineIn Status Disabled	LineIn Level 1.33 Volts p-p	LineOut Level 2.83 Volts p-p	Phone Volume 80 %
LineIn→Delay 0 %	LineIn→Reverb 100 %	Unassigned 0	Unassigned 0
1	1	1	1

Preset 1: Dohee FM1 WavPlyr  RawPlyr  Synth  FmSynth  LineIn 

MIDI Settings and Controllers

MIDI Settings

Future Development

 Save Changes  Copy Settings 

MIDI Port USB MIDI	MIDI In Device LaunchKey25	Wav Player Chan 1	Raw Player Chan 1
BasicSynth Chan 1	FmSynth Chan 1	Unassigned 0	Unassigned 0
Unassigned 0	Unassigned 0	Unassigned 0	Unassigned 0
Unassigned 0	Unassigned 0	Unassigned 0	Unassigned 0
1	1	1	1

Preset 1: Dohee FM1 WavPlyr  RawPlyr  Synth  FmSynth  LineIn 

Supprt Screens

Presets Live

Presets Live

Direct Preset & Bank Access

All Notes Off

Ignore Note Off

1

Dohee FM1

2

Synth 1

3

Synth 2

4

Synth Rpt

5

Dohee Efx

6

Synth Rvb

7

Synth Rich

8

Woo Woo

9

MicroTone L

10

Synth Pulse

11

MicroTone S

12

Synth Marim

13

WAV Plyr

14

Raw Plyr

15

Basic Synth

16

FM Cluster

17

FM Synth

18

Dohee FM2

19

Dohee Shift

20

Multi 1

21

22

Synth Rpt 2

23

Guitar

24

25

Synth Pulse

26

Cluster

27

FM Piano

28

29

30

31

32

ALL OFF

DefaultBank.pbmk

DefaultBank-250220.pbmk
DefaultBank-250319.pbmk
DefaultBank-250415.pbmk
DefaultBank-250519.pbmk
ImportActiveTest.pbmk
SynthMicroToneCmnDly.pbmk
FmSynthPanTest.pbmk
MultiPlayerTest1.pbmk
MultiPlayerTest2.pbmk
WavPlayerMultiEffects.pbmk
BasicSynthMultiEffects.pbmk

Active Bank: DefaultBank.pbmk

Presets on SD

Presets 1: Dohee FM1

WavPlyr RawPlyr Synth FmSynth LineIn

Presets Manager

Presets Management

Load,Save,Edit Presets & Banks

1

Dohee FM1

2

Synth 1

3

Synth 2

4

Synth Rpt

5

Dohee Efx

6

Synth Rvb

7

Synth Rich

8

Woo Woo

9

MicroTone L

10

Synth Pulse

11

MicroTone S

12

Synth Marim

13

WAV Plyr

14

RAW Plyr

15

Basic Synth

16

FM Cluster

17

FM Synth

18

Dohee FM2

19

Dohee Shift

20

Multi 1

21

22

Synth Rpt 2

23

Guitar

24

25

Synth Pulse

26

Cluster

27

FM Piano

28

29

30

31

32

ALL OFF

Presets: MicroTone L

Active Bank: DefaultBank

Press Load, Save, Import, Clear, or Rename

Press Load or Save

Load

Save

Import

Clear

Rename

Cancel

Load

Save

Default

SD Card

Cancel

Presets Management

Load, Save, Edit Presets & Banks



Dohee FM1	Synth 1	Synth 2	Synth Rpt	Dohee Efx	Synth Rvb	Synth Rich	Woo Woo
1	2	3	4	5	6	7	8
MicroTone L	Synth Pulse	MicroTone S	Synth Marim	WAV Plyr	RAW Plyr	Basic Synth	FM Cluster
9	10	11	12	13	14	15	16
FM Synth	Dohee FM2	Dohee Shift	Multi 1		Synth Rpt 2	Guitar	
17	18	19	20	21	22	23	24
Synth Pulse	Cluster	FM Piano					ALL OFF
25	26	27	28	29	30	31	32

Preset: MicroTone L

Active Bank: DefaultBank

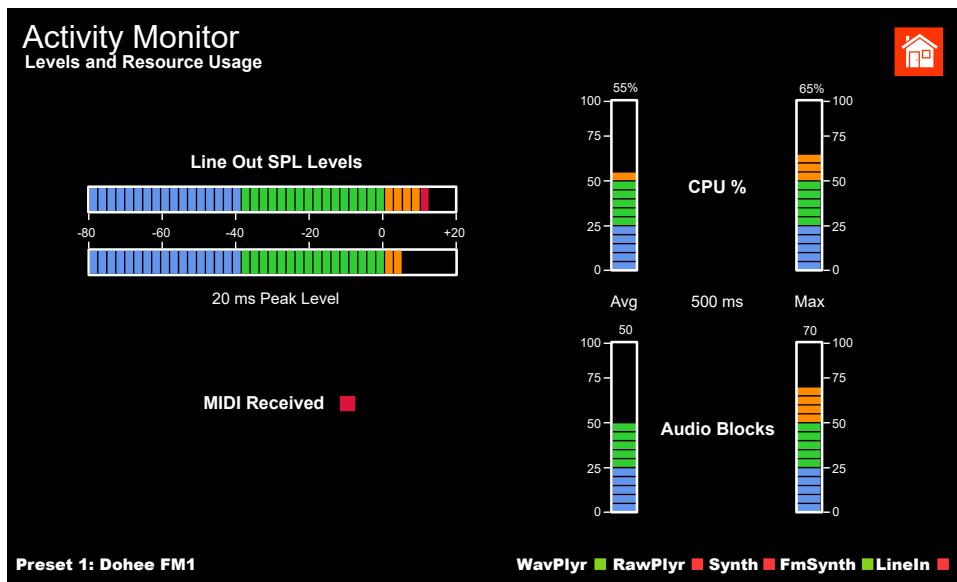
Press Load, Save, Import, Clear, or Rename

Press Load or Save

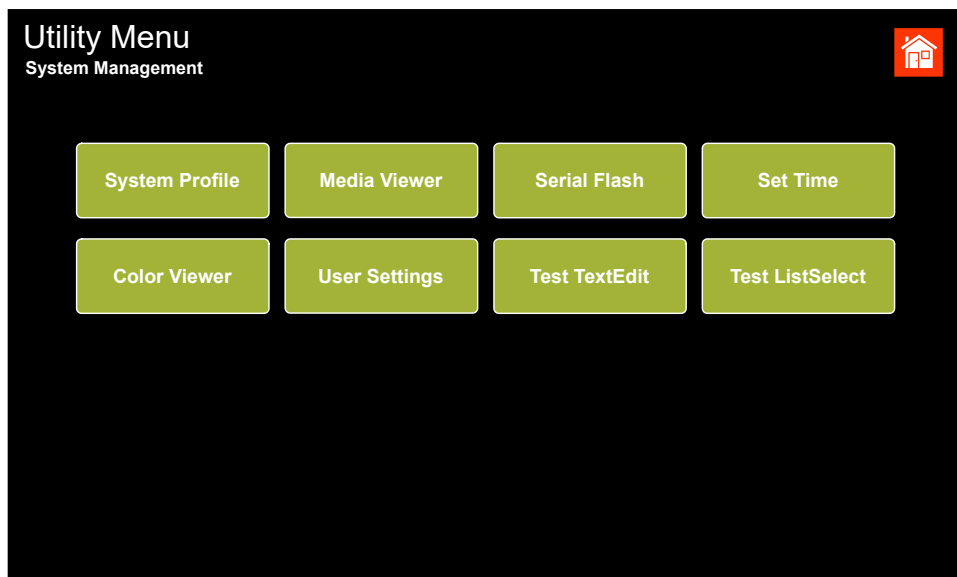
Load Save Import Clear Rename Cancel

Load Save Default SD Card Cancel

Activity Monitor



Utility Menu



Appendix

References

AudioPlatform Source and Documentation

[AudioPlatform Source Code](#)

[AudioPlatform Documentation](#)

ProtoSupplies

[ProtoSupplies Web Site](#)

[Prototyping System For Teensy 4.1](#)

[Teensy 4.1 Fully Loaded For Prototyping System](#)

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PJRC / Teensy

[PJRC](#)

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Teensy DIY Audio Projects and Kits

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[TeensyMix Synth - DIY 8 Voice Synthesizer](#)

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[MicroDexed Touch Synthesizer](#)

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YouTube

[DIY Dirtywave M8 headless using a Teensy 4.1](#)

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